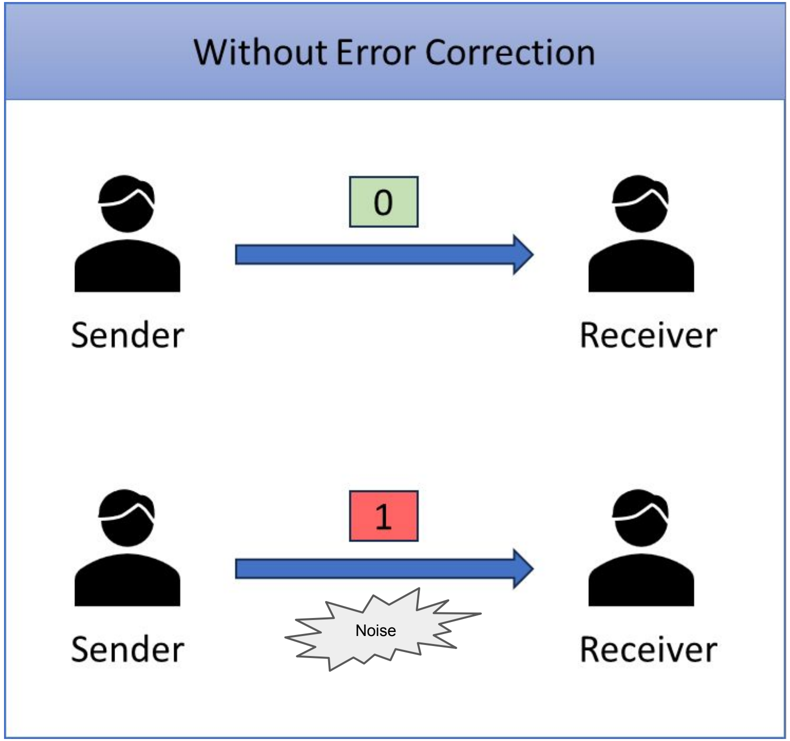
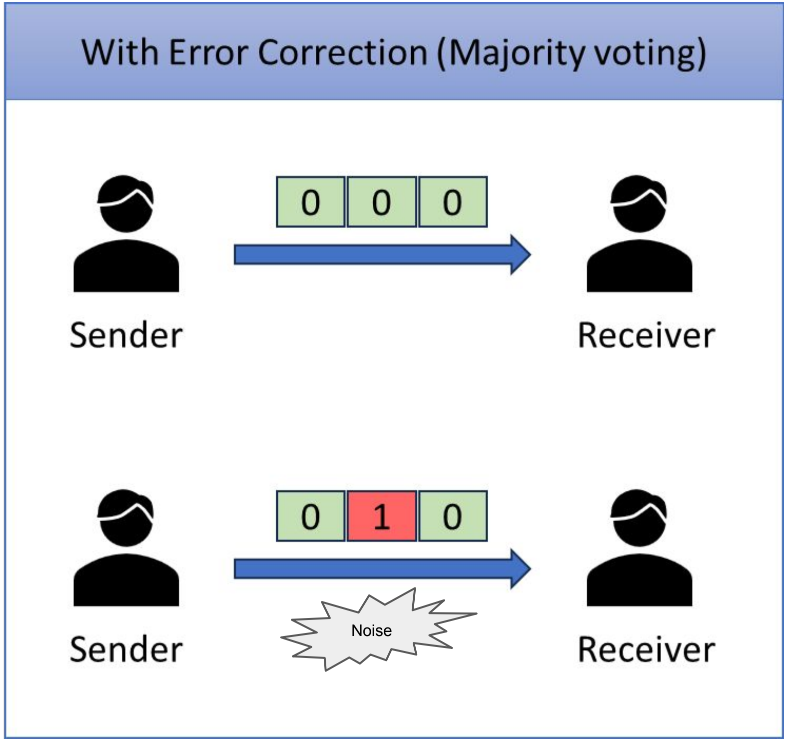
Topological Qubit Error Correction on a Donut

Challenge #7

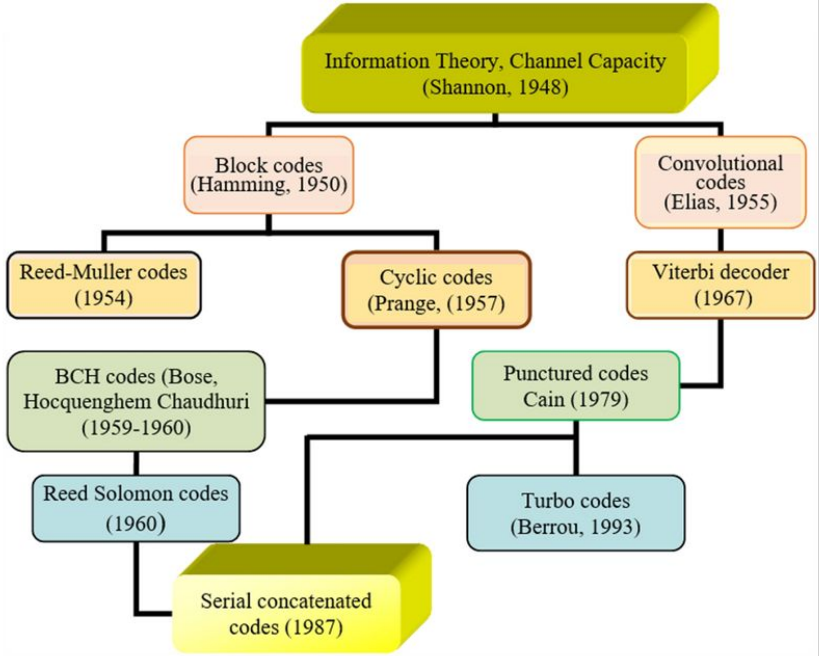
By: Leif, Rose, Vijay and Gehad

Classical error correction





Classical error correction types

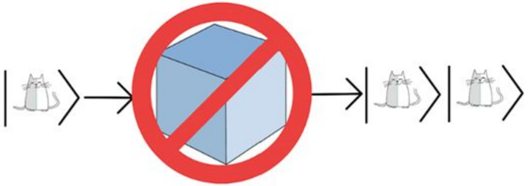


Application	Types	Criteria
Network	Checksum, CRC	Speed, Low Latency
Storage	Reed-Solomon, Hamming code	High Reliability
Satellite	Convolutional, Turbo codes	High Noise

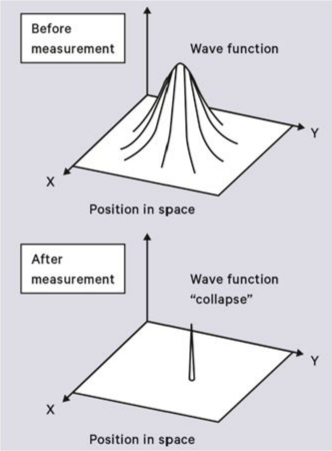
History summary relating to the ECC | Download Scientific Diagram

Classical to Quantum error correction

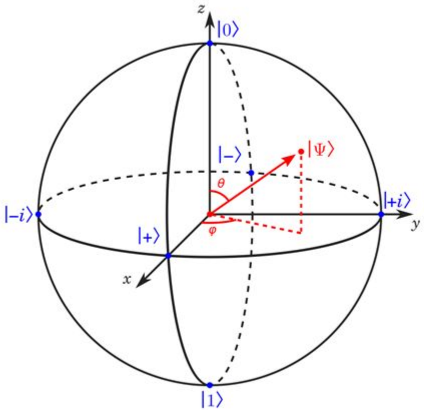
Problem1: No Cloning Theorem



Problem2: Wavefunction collapse

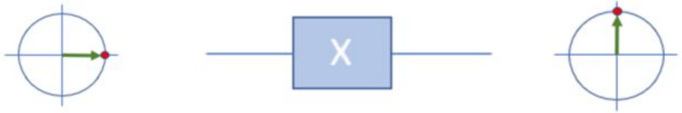


Problem3: Continuous state

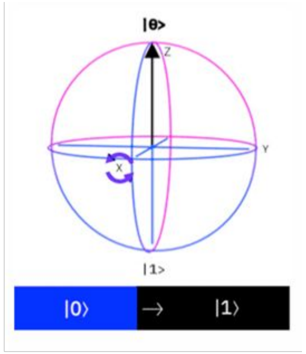


What does quantum state error looks like?

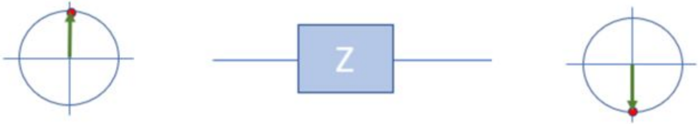
A. Simulating bit flip using X gates



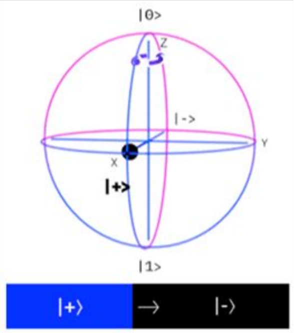
$\psi = 1.0|0\rangle + 0.0|1\rangle$ $\quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad$ $\psi = 0.0|0\rangle + 1.0|1\rangle$



B. Simulating phase flip using Z gates



$\psi = 0.0|0\rangle + 1.0|1\rangle$ $\quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad$ $\psi = 0.0|0\rangle - 1.0|1\rangle$



Quantum Computing | ShareTechnote

Quantum Error Correction Workflow

Initial State

Noise

Errored State

Error Correction

Initial State
(hopefully)

[A Methods Focused Guide to Quantum Error Correction and Fault-Tolerant Quantum Computation | Quantum repetition code for bit-flips]

Stabilizer Code

Goal

Correct X, Y, Z errors

Stabilizers

They don't change the eigenvalues of the state

$$S_i|\psi\rangle = |\psi\rangle \quad \forall i$$

Encode Qubits

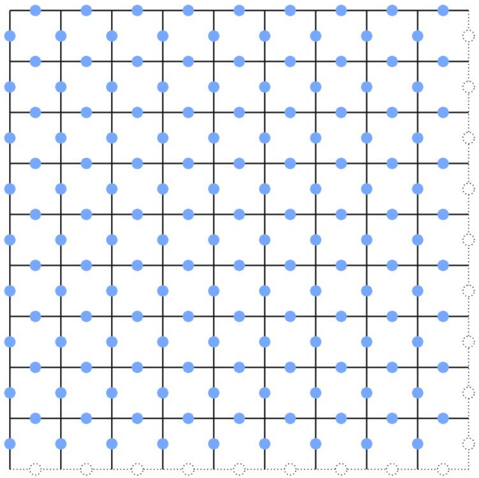
The encoding depends on the choice of stabilizers

Errors

If E anti-commutes with a specific stabilizer S_i

$$S_i(E|\psi\rangle) = -(E|\psi\rangle)$$

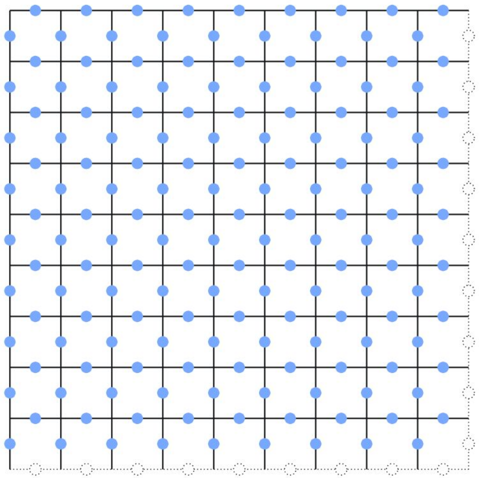
Surface/ donut code



→ encodes **2 logical qubits**

Toric code | IBM Quantum Learning

Surface/ donut code

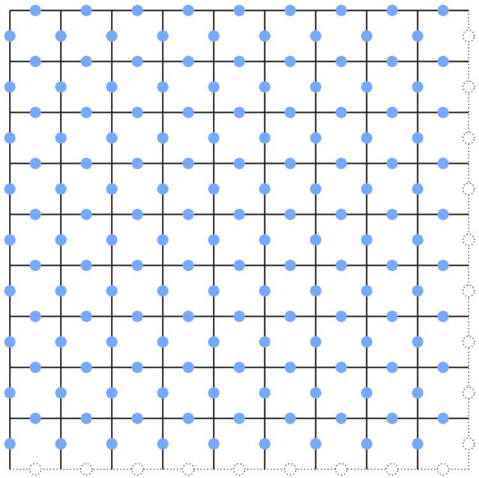


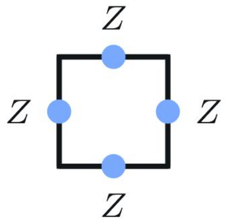
+ periodic boundary conditions

→ encodes **2 logical qubits**

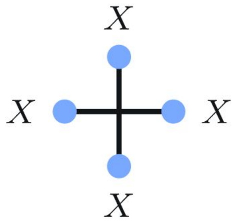
Toric code | IBM Quantum Learning

Surface/ toric code





Z stabilizer



X stabilizer

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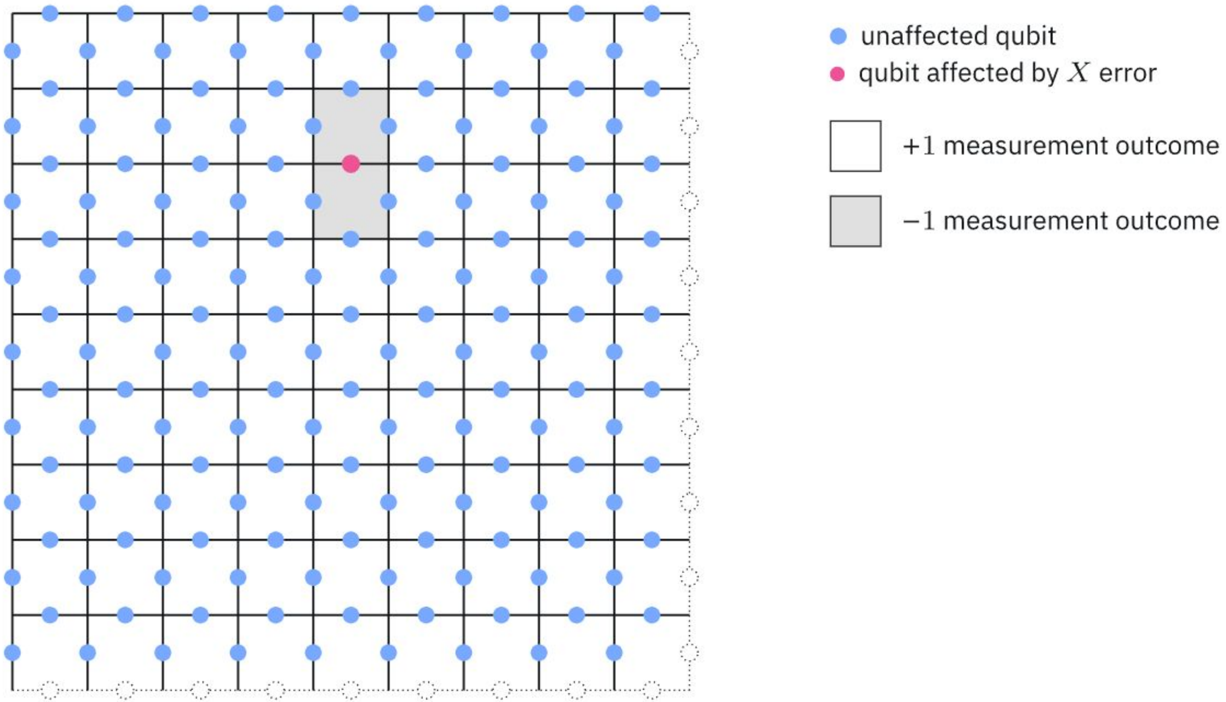
Detecting Errors



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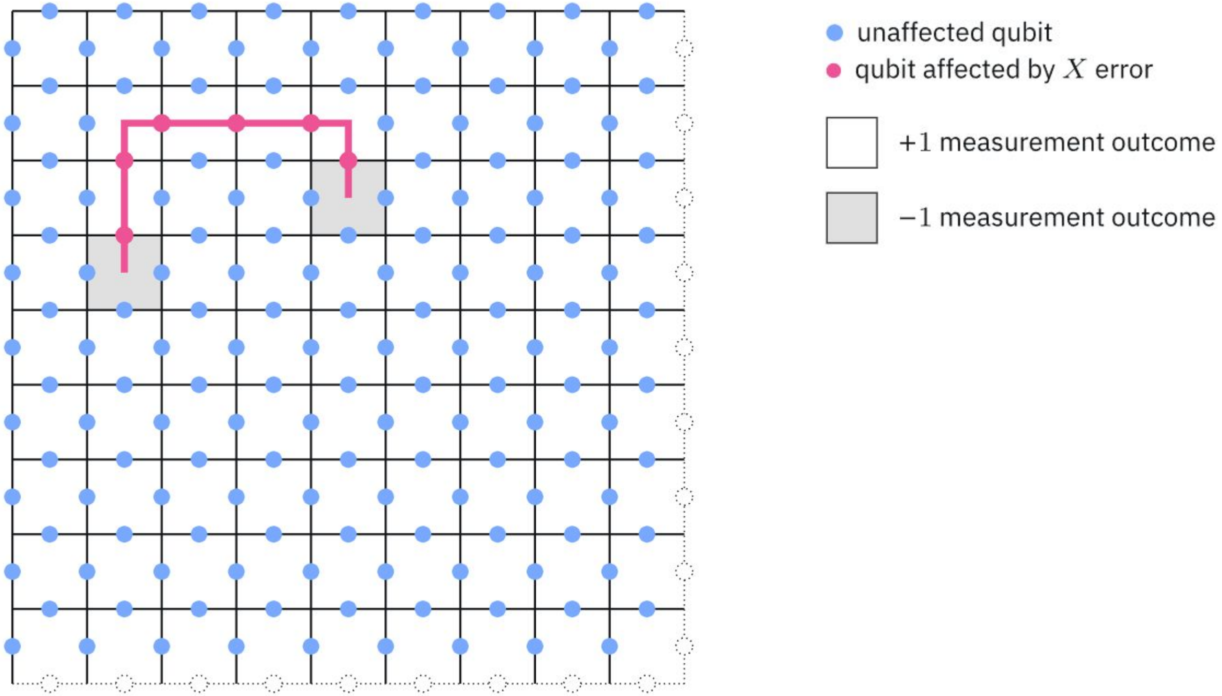
Detecting Errors



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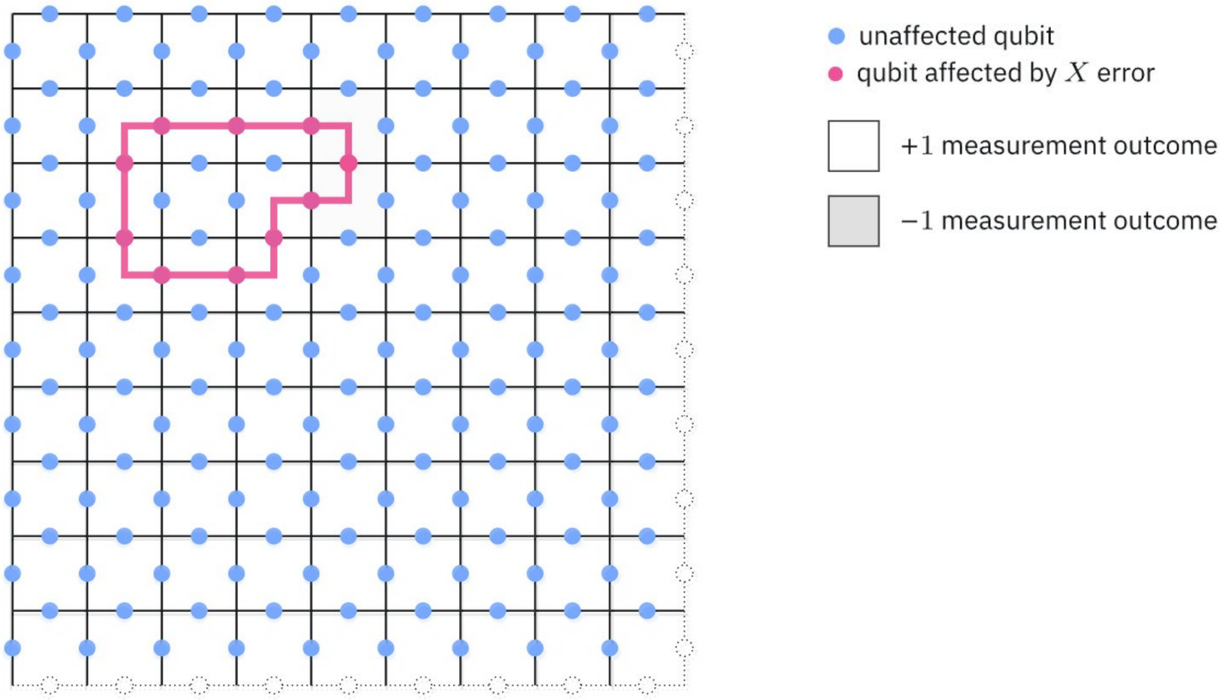
Detecting Errors



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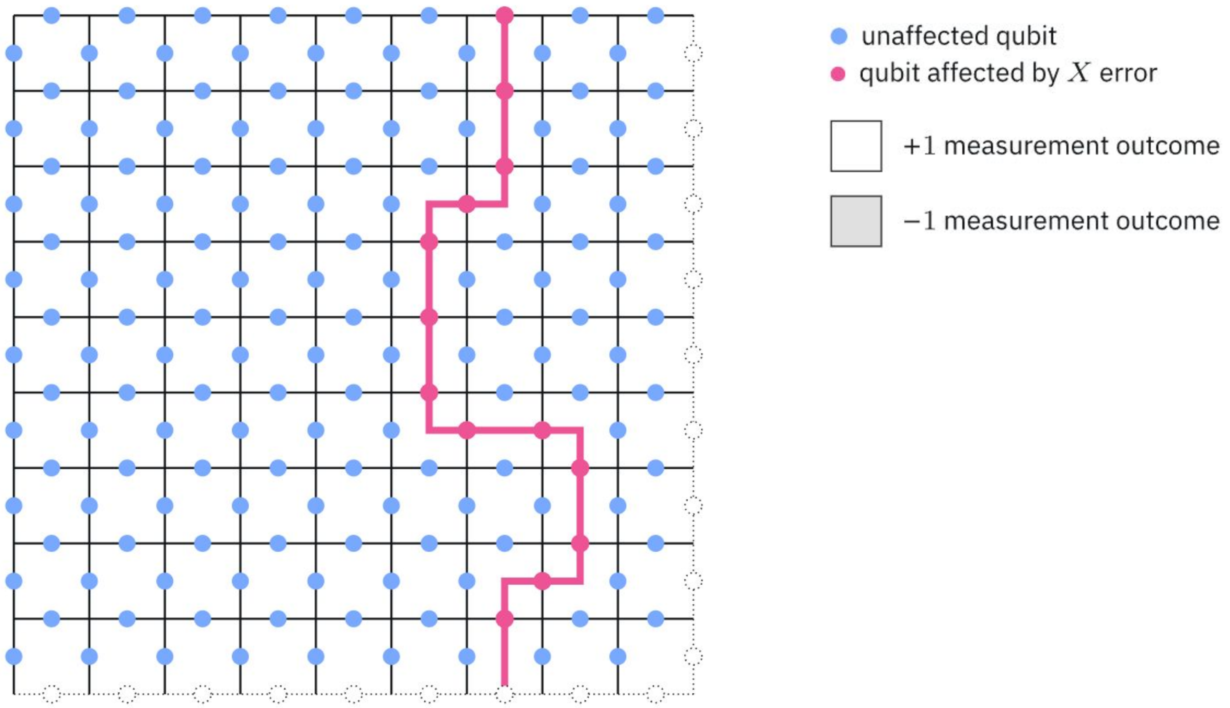
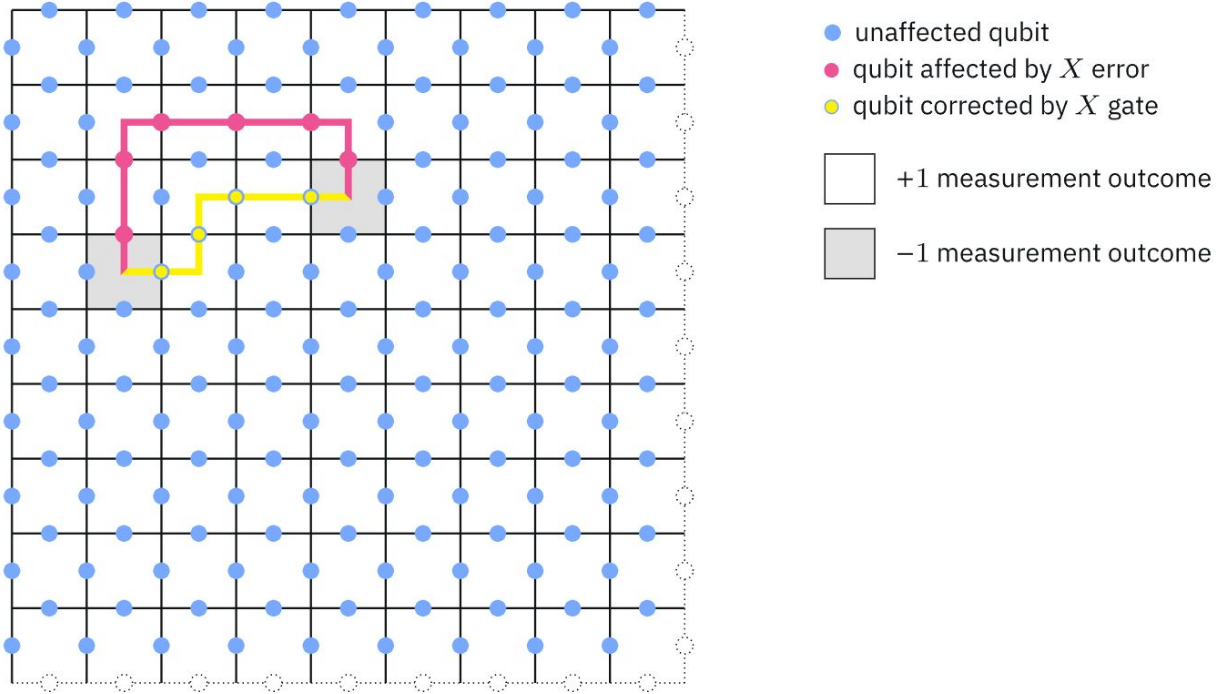
Detecting Errors



[Toric code | IBM Quantum Learning](#)

Correcting Errors



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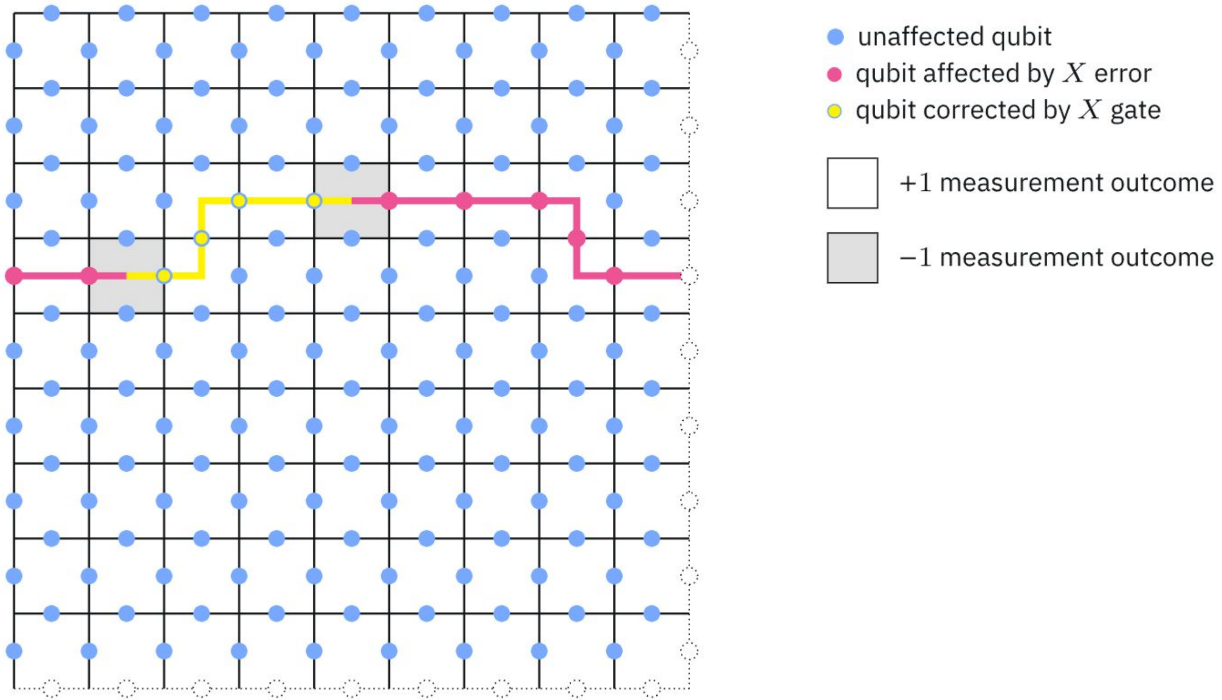
Correcting Errors



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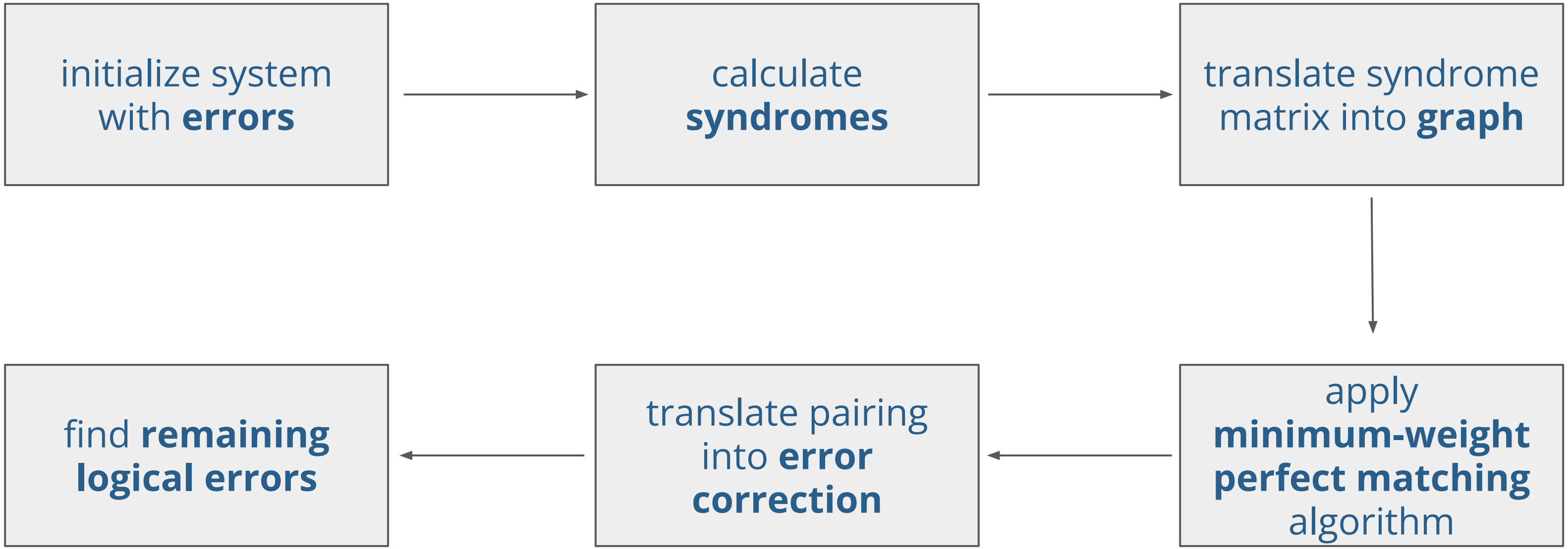
Simulation workflow

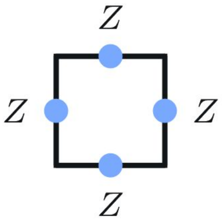


Simulation workflow

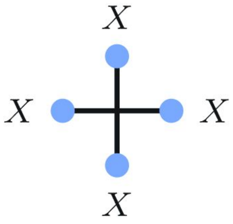
initialize system
with **errors**

The “bases” of our stabilizers/flips

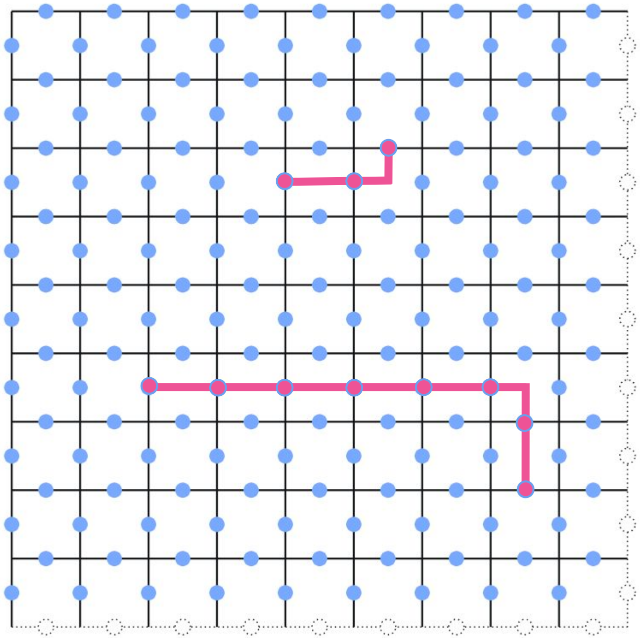
$$\begin{bmatrix} X & + & X & + \\ & X & & X \\ X & + & X & + \\ & X & & X \end{bmatrix} \qquad \begin{bmatrix} Z & + & Z & + \\ & Z & & Z \\ Z & + & Z & + \\ & Z & & Z \end{bmatrix}$$



Z stabilizer



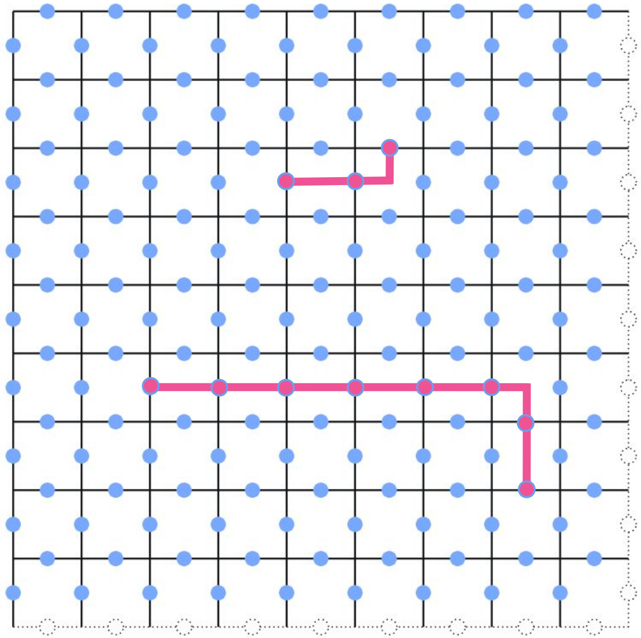
X stabilizer

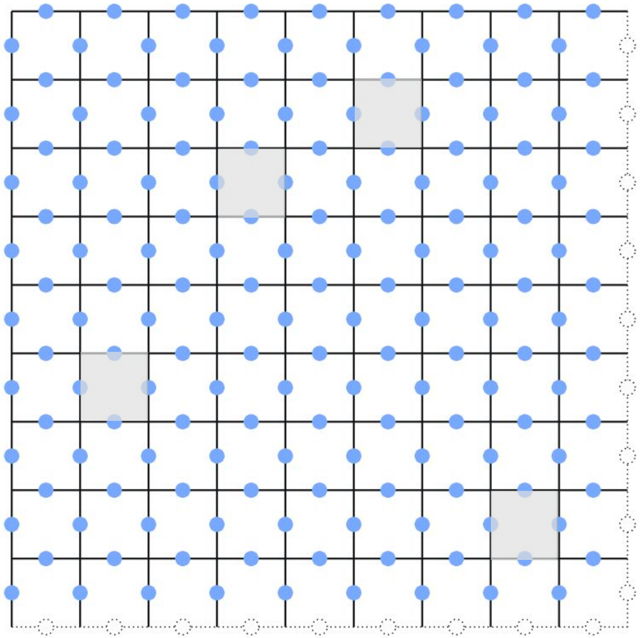


errors

Simulation workflow

initialize system with **errors** → calculate **syndromes**



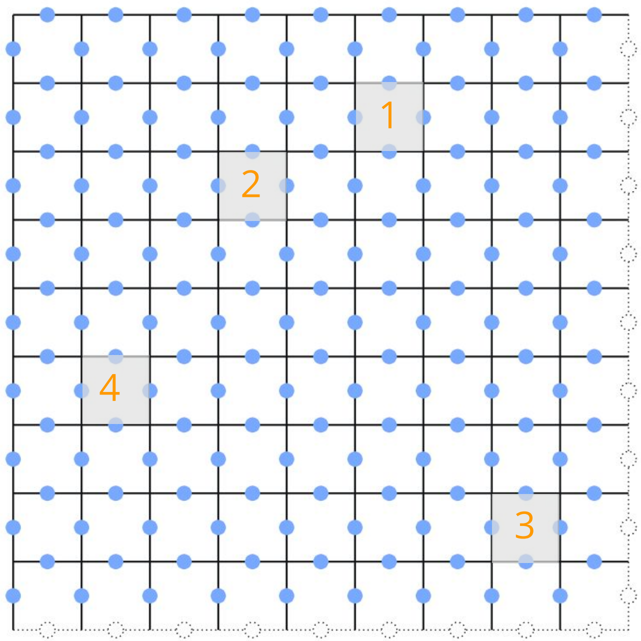


errors

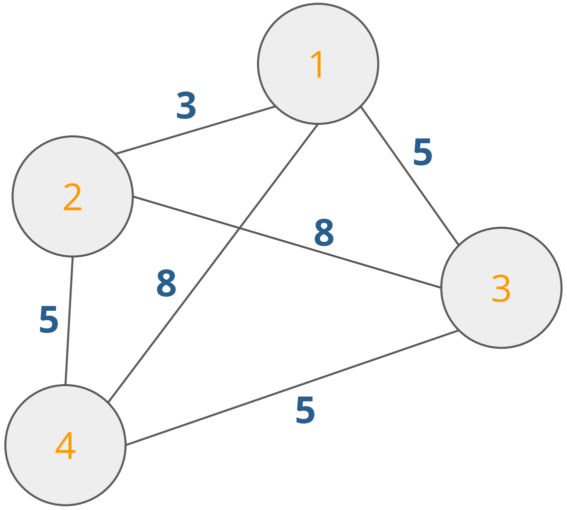
syndromes

Simulation workflow

initialize system with **errors** → calculate **syndromes** → translate syndrome matrix into **graph**



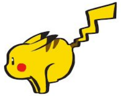
syndromes

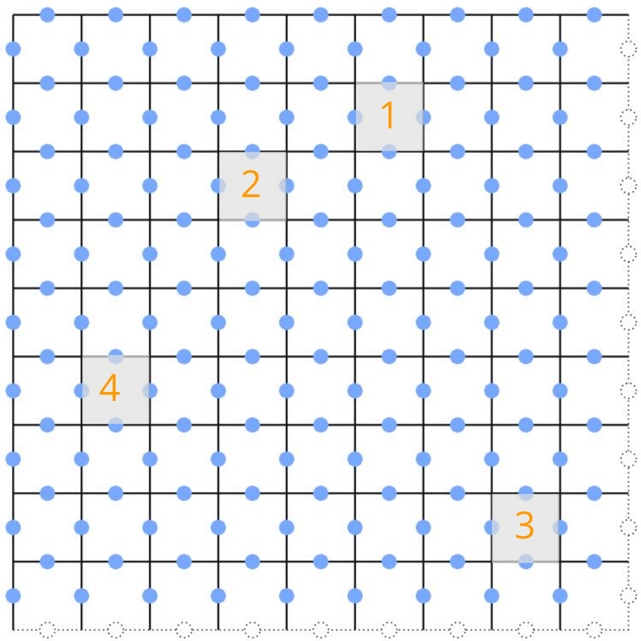


graph

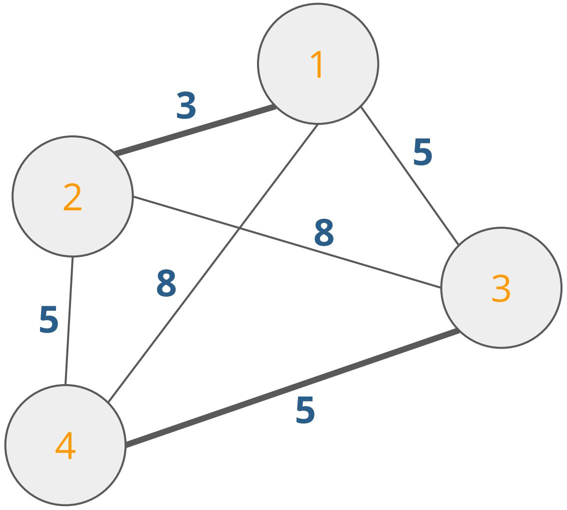
Simulation workflow

initialize system with **errors** → calculate **syndromes** → translate syndrome matrix into **graph** → apply **minimum-weight perfect matching** algorithm





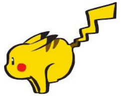
syndromes

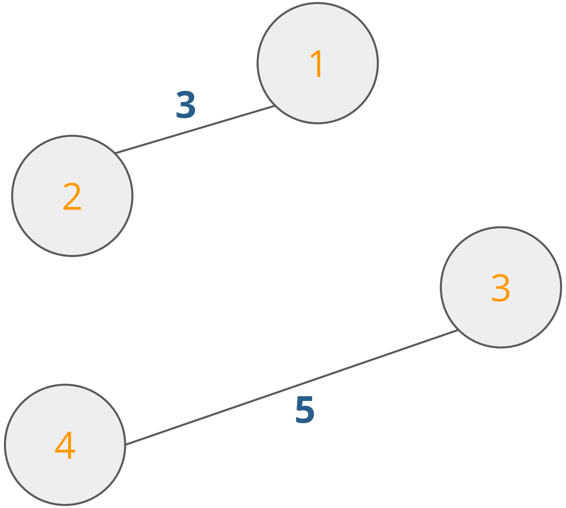


graph

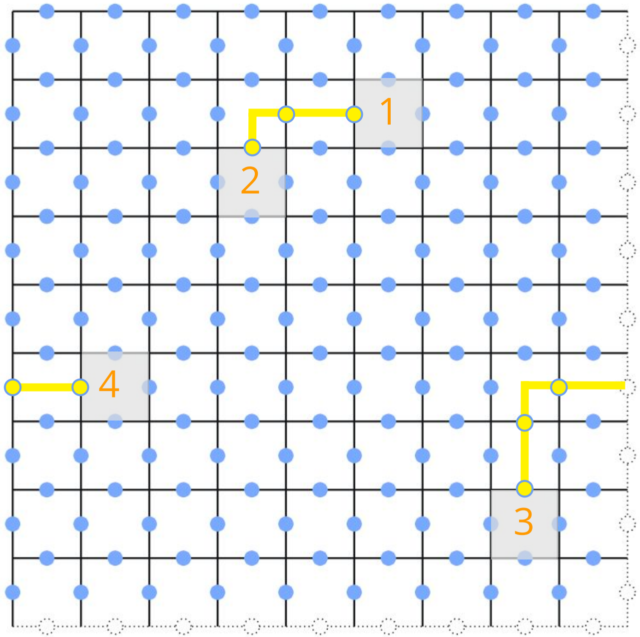
Simulation workflow

initialize system with **errors** → calculate **syndromes** → translate syndrome matrix into **graph** → apply **minimum-weight perfect matching** algorithm → translate pairing into **error correction**





minimum-weight perfect matching



error correction

Simulation workflow

initialize system with **errors**

calculate **syndromes**

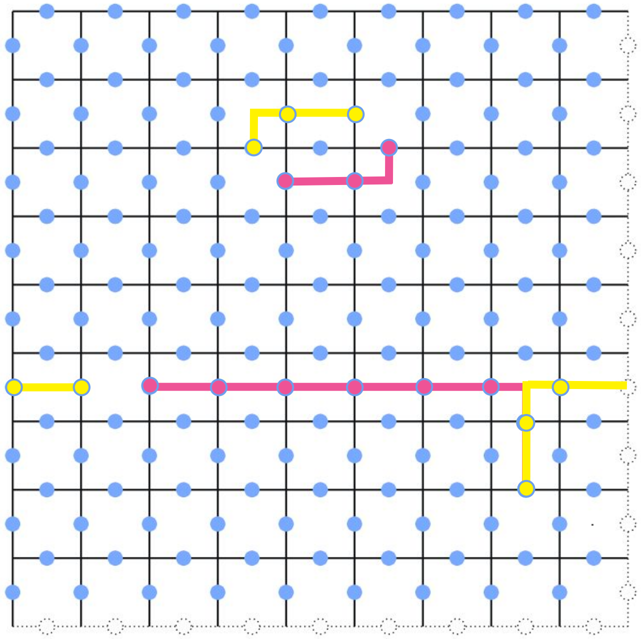
translate syndrome matrix into **graph**

apply **minimum-weight perfect matching** algorithm

translate pairing into **error correction**

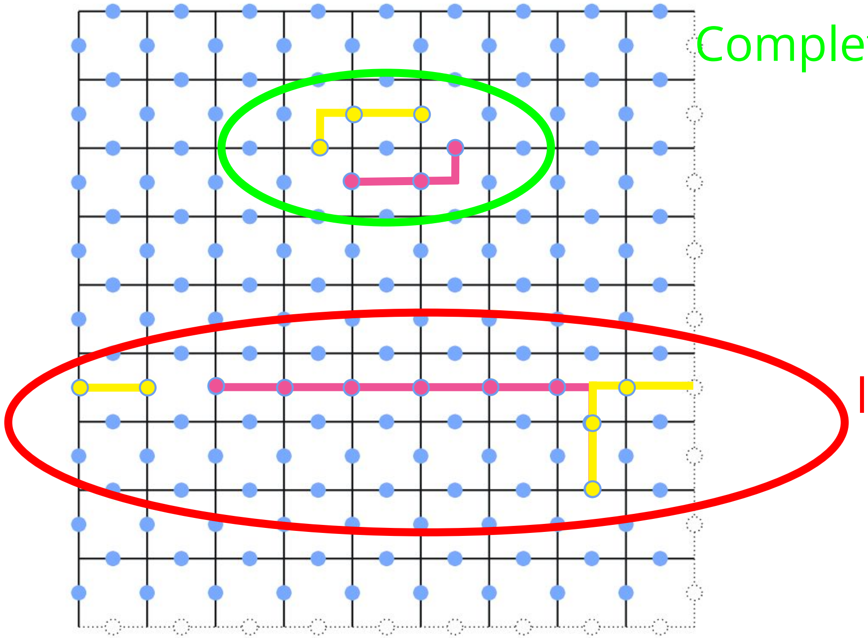
find **remaining logical errors**

Detecting Logical Errors



errors + error correction

Detecting Logical Errors



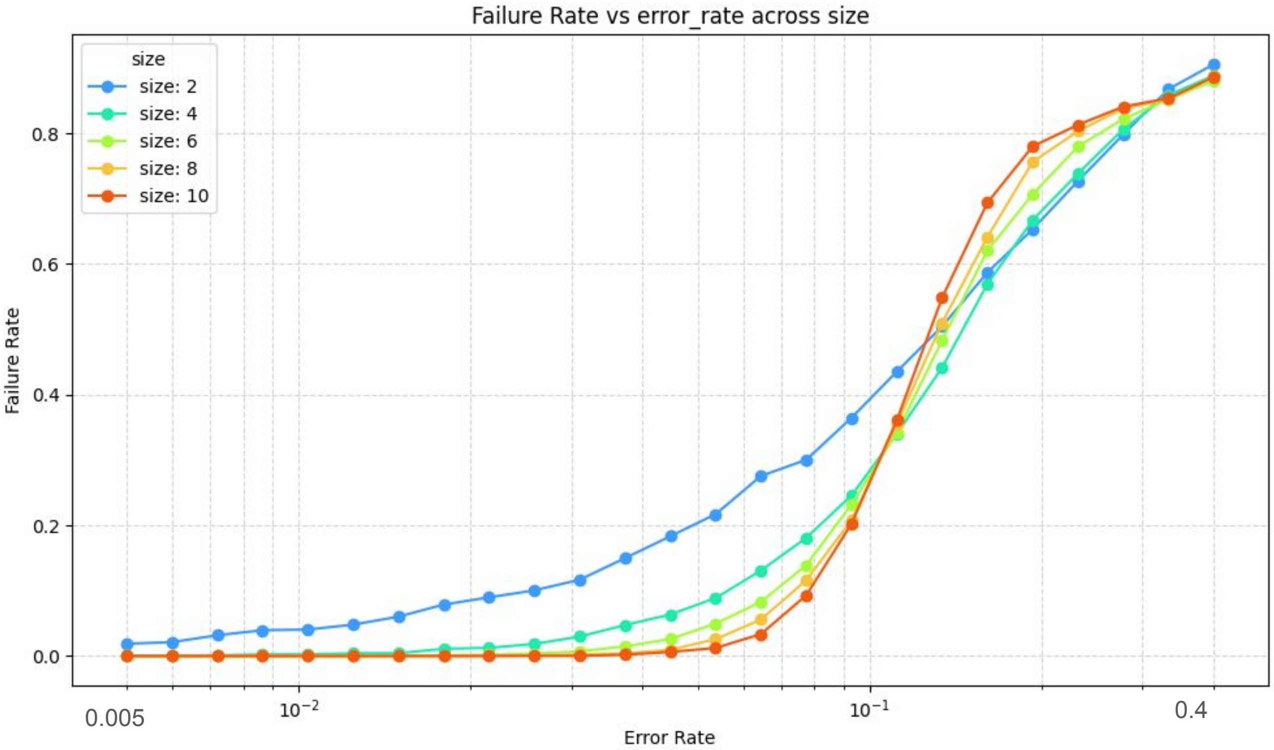
Completely fine

logical error

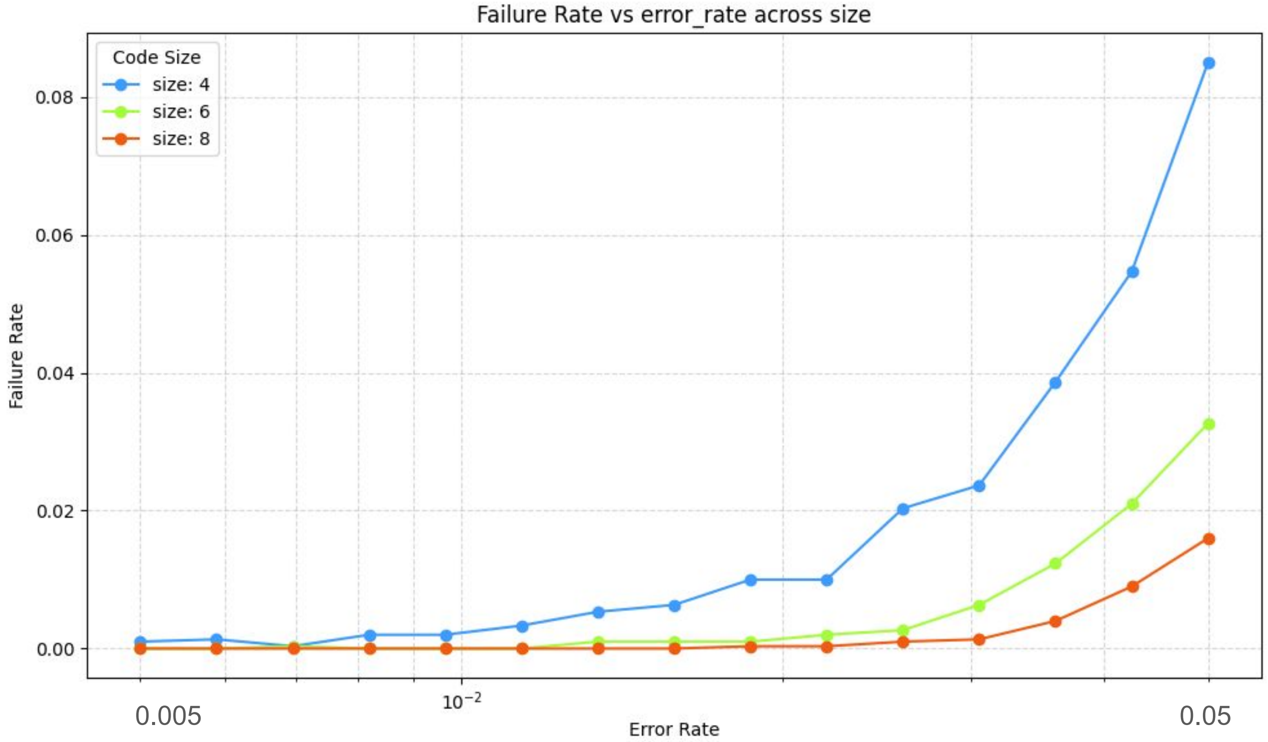
errors + error correction

Error Rate Threshold

- Threshold around 1.2%
- Will use size 6 due to time constraints
- Algorithm scales with O(n^3log(n)):

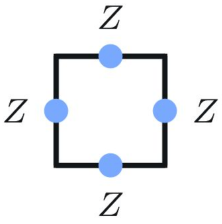


Error Rate Convergence Relative to Size

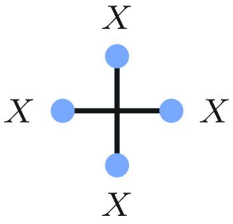


The “bases” of our stabilizers/flips

$$\begin{bmatrix} X & + & X & + \\ & X & & X \\ X & + & X & + \\ & X & & X \end{bmatrix} \qquad \begin{bmatrix} Z & + & Z & + \\ & Z & & Z \\ Z & + & Z & + \\ & Z & & Z \end{bmatrix}$$



Z stabilizer

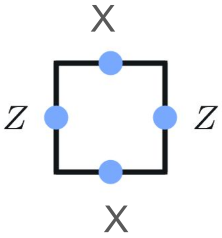


X stabilizer

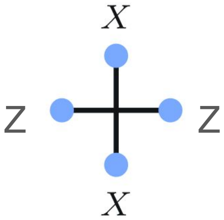
Toric code | IBM Quantum Learning

Our new stabilizers

$$\begin{bmatrix} Z & + & Z & + \\ & X & & X \\ Z & + & Z & + \\ & X & & X \end{bmatrix} \qquad \begin{bmatrix} X & + & X & + \\ & Z & & Z \\ X & + & X & + \\ & Z & & Z \end{bmatrix}$$



Z stabilizer



X stabilizer

[Toric code | IBM Quantum Learning](#)

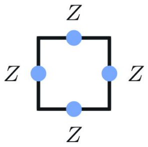
The Programming Trick

$$\begin{bmatrix} X & + & X & + \\ & X & & X \\ X & + & X & + \\ & X & & X \end{bmatrix} \quad \begin{bmatrix} Z & + & Z & + \\ & Z & & Z \\ Z & + & Z & + \\ & Z & & Z \end{bmatrix}$$

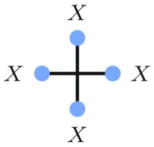
Flips Input 1 ↓ Flips Input 2

↓

Failure rates



Z stabilizer



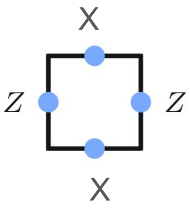
X stabilizer

$$\begin{bmatrix} Z & + & Z & + \\ & X & & X \\ Z & + & Z & + \\ & X & & X \end{bmatrix} \begin{bmatrix} X & + & X & + \\ & Z & & Z \\ X & + & X & + \\ & Z & & Z \end{bmatrix}$$

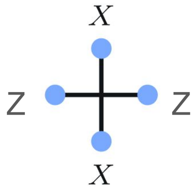
Flips Input 1 ↓ Flips Input 2

↓

Failure rates in new basis

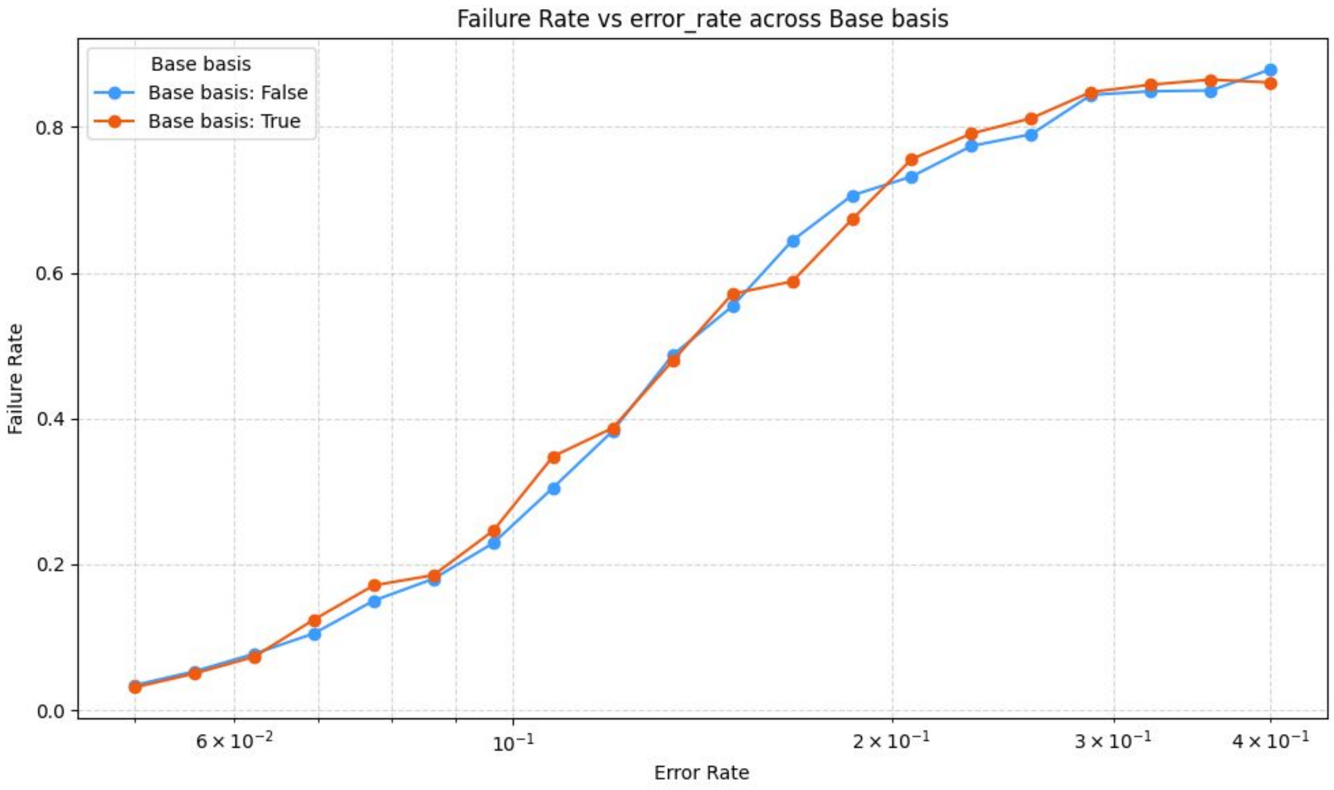


Z stabilizer

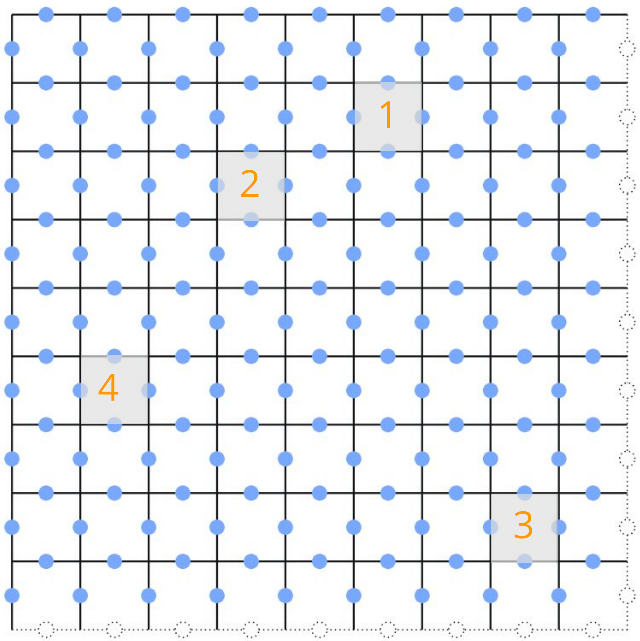


X stabilizer

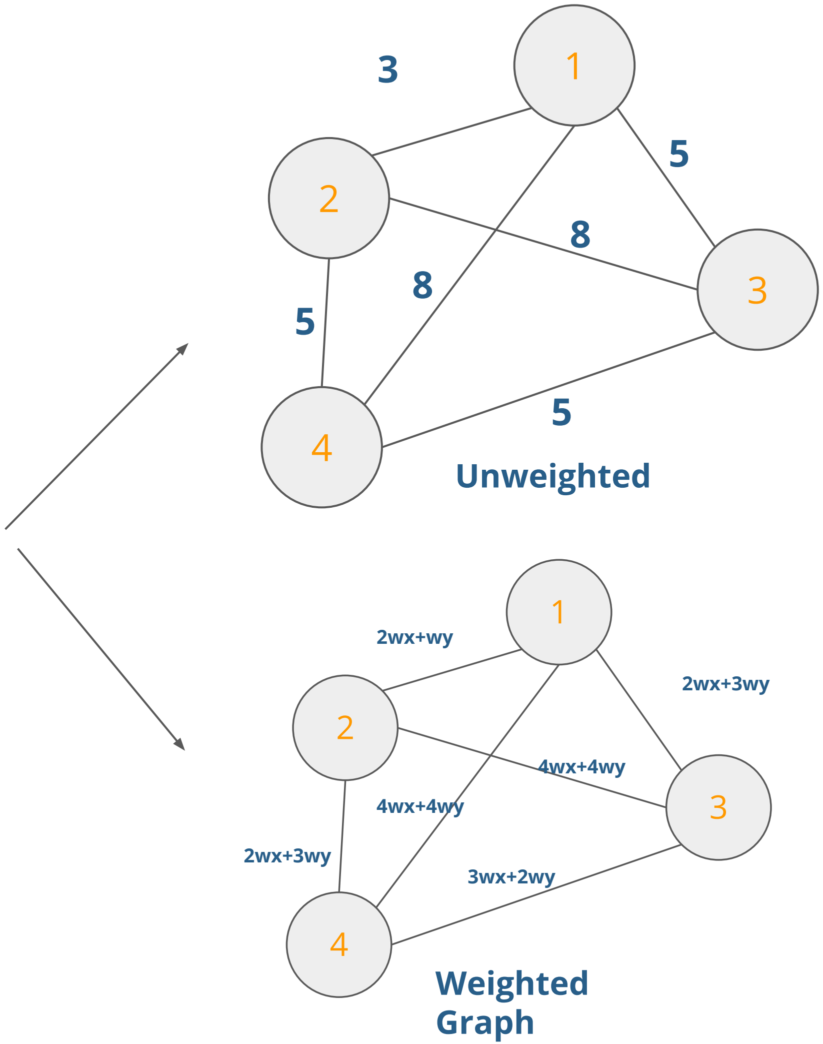
Models Similarities



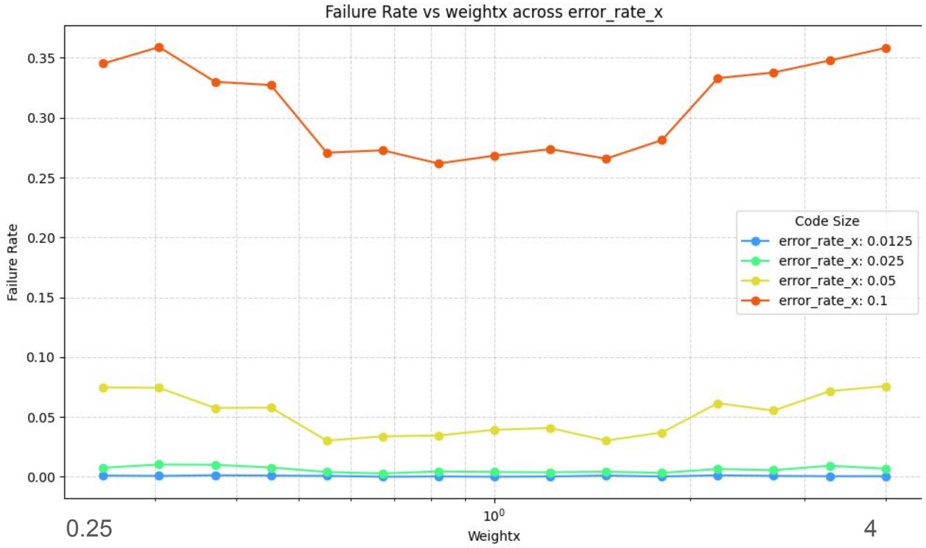
Weights



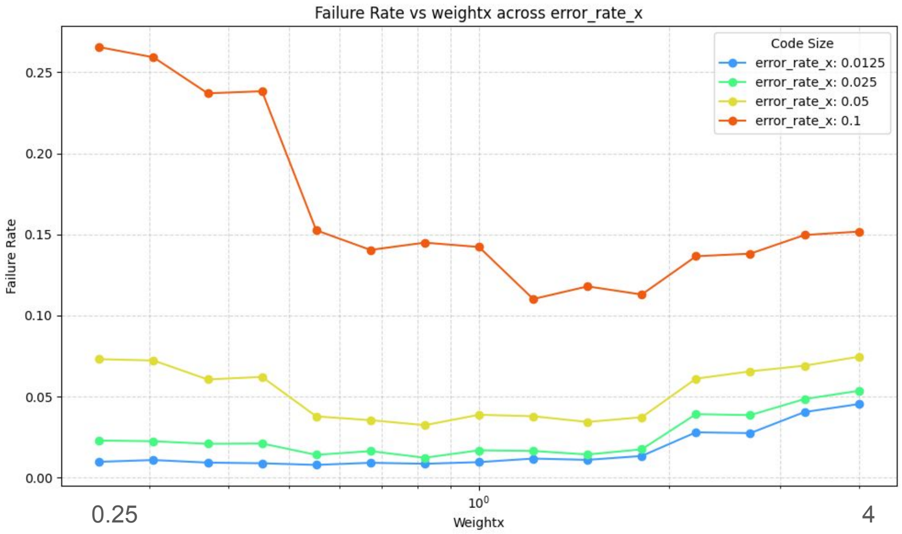
Weight = dx + dy →dx*wx + dy*wy



Comparing with Weights



XXXX,ZZZZ syndromes



XZZX syndromes

Thank you for Listening!

And shoutout to Elena our supervisor!

And a shoutout to our sponsors! :)

DUNKIN' DONUTS® DigiQ DIGITALLY ENHANCED QUANTUM TECHNOLOGY MASTER

References

- Quantum Computing | ShareTechnote
- Single Qubit Gates | Learning Quantum

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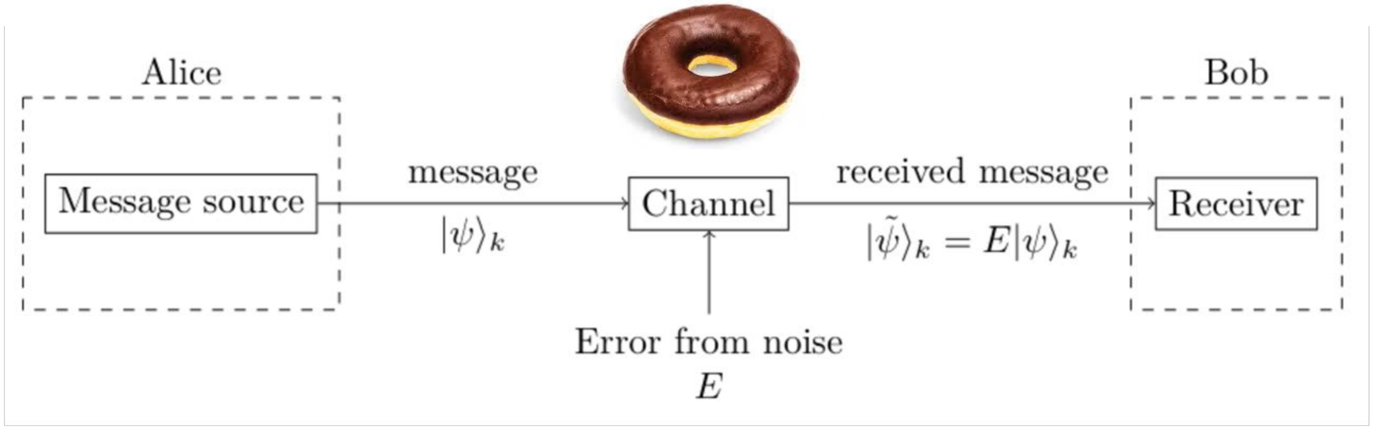
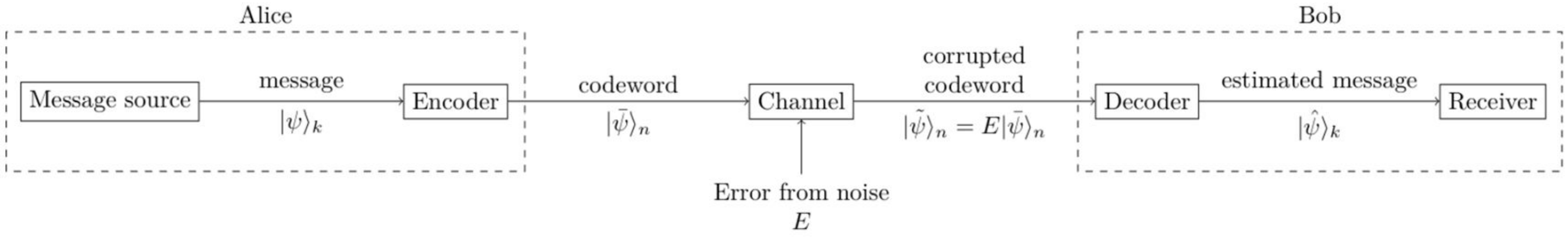
Quantum Error Correction Workflow





[A Methods Focused Guide to Quantum Error Correction and Fault-Tolerant Quantum Computation | Quantum repetition code for bit-flips]